SHREYA VENKATESH

Email ● Personal Website ● LinkedIn ● Google Scholar ● GitHub

BIOMEDICAL SCIENTIST | ENGINEER | SCIENCE COMMUNICATOR

Interdisciplinary biomedical scientist with expertise spanning cancer biology, aging, mechanobiology, and translational research. Experienced in experimental and computational research, scientific and policy writing, science communication, stakeholder engagement, and interdisciplinary collaboration.

EDUCATION

University of Colorado Boulder – M.S., Ph.D., Mechanical Engineering | 2023, 2026
Dissertation: Mechanical Loading Governs Osteocyte-Tumor Cell Signaling in Metastatic Bone Disease

Cornell University – B.S., Biomedical Engineering | 2021

RESEARCH PUBLICATIONS

2026 (In preparation) **Venkatesh S**, Ferguson VL, Lynch ME. "Advanced glycation end-product accumulation in the onco-aging environment as a driver of bone metastatic breast cancer." *Review*.

2026 (In preparation) **Venkatesh S**, Thompson WR, Lynch ME. "Mechanically loaded breast cancer cells prevent sclerostin downregulation by mature osteocytes in a co-culture model of the metastatic vicious cycle."

2026 **Venkatesh S**, et al. "Radiation modulates the mechanoresponse of bone-homing triple-negative breast cancer cells." *Tissue Engineering Part A*, *Mechanobiology in Tissue Engineering Special Issue*.

2025 **Venkatesh S**, et al. "High-fidelity computational fluid dynamics modeling to simulate perfusion through a bone-mimicking scaffold." *Computers in Biology and Medicine*.

2024 Lang BJ, Holton KM, Guerrero-Gimenez ME, Okusha Y, Magahis PT, Shi A, Neguse M, **Venkatesh S**, et al. "Heat shock protein 72 supports extracellular matrix production in metastatic mammary tumors." *Cell Stress and Chaperones*.

OTHER PUBLICATIONS & GRANT-WRITING EXPERIENCE

2026 **Venkatesh S**, Groves AR, Joshi P (co-first authors). "Addressing Elderly Loneliness with Artificial Intelligence Companionship." *MIT Science Policy Review, Volume VII*.

2026 **Venkatesh S**, Affsprung D, Mattson A. "The Past Is Our Present: A History of America's Complicated Relationship with Science." *Science Policy in a SNAP.*

2025 **National Institute of Aging (NIA) R36 Dissertation Awards**, PAR-24-052. "Aging-induced AGE accumulation heightens bone fragility in metastasis by dysregulating osteocytes." Total: $140,000. Impact Score: 30. (grant decisions suspended)

2023 Hayes A, Sundaram V, Williams G, **Venkatesh S**, et al. "Meeting Schools Where They Are: Integrating Engineering Outreach Curriculum in the Classroom Without Forcing an Agenda." *American Society for Engineering Education: Rocky Mountain Section Conference*.

RESEARCH & ENGINEERING PROJECTS

Aug 2021 – 2026 **Cancer and Bone Doctoral Researcher**
PI: Dr. Maureen E. Lynch *University of Colorado Boulder*

- Characterized breast cancer cell responses to fluid shear stress in a novel 3D non-animal model, informing targets for anti-tumorigenic and bone-targeted anti-resorptive therapies.
- Generated computer simulations of fluid shear stress distributions within bone-mimicking scaffold models.
- Spearheaded the team's development of onco-aging bone models and integration of next-generation sequencing methods, from experimental pipeline to computational analyses via supercomputing clusters.
- Managed collaborations with clinical and pre-clinical researchers across various disciplines (e.g., pathology, genomics, data science, orthopedics, fluid mechanics).
- Demonstrated proficiency at distilling complex scientific information into compelling takeaways for expert and lay audiences (3-Minute Thesis Finalist, Best Poster/Oral Presentation Awards, Young Investigator Travel Awards, public talks at the Denver Museum of Natural History).

- Published impactful peer-reviewed research output in Computers in Biology and Medicine, Tissue Engineering Part A.
- Strong track record of departmental service as a graduate student representative, K-12 outreach program lead, and community event organizer, resulting in department- and university-wide recognition.

Feb – May 2021 **Biomechanical Data & Testing Industry Liaison**
Mentor: Dr. Lawrence Bonassar *Cornell University & MTF Biologics*

- Served as scientific liaison for a university-industry partnership, designing and executing mechanical testing of acellular dermal matrices used in breast reconstruction.
- Analyzed experimental data and translated findings into performance evaluations and recommendations for MTF Biologics scientists, supporting product integration and safety analysis.

May – Aug 2020 **Entrepreneurship Bootcamp Software Lead**
Technical & Product Development Teams *ORIGIN Entrepreneurship Program*

- Developed a business concept and software prototype to reduce local food waste, contributing to web development, product design, and a pitch to an external business panel.

Jan – May 2020 **Biomechanical Data Analyst**
Mentor: Dr. Jonathan Butcher *Cornell University*

- Modeled cardiovascular pressure and flow using Excel and CircAdapt and analyzed motion-capture and force-plate data using MATLAB and OpenSim.

Jan 2019 – May 2021 **Engineers Without Borders (EWB) Team Technical Lead**
Project Team *Cornell Aerial Robotics Team*

- Designed and assembled Tarot 680PRO IRONMAN quadcopter for testing - aided in motor testing, speed controller modification, servo connections, and stability modifications.
- Led EWB team through assembly, testing, and design of quadcopter for use in digital agriculture.

Jan 2019 – May 2021 **Tendon Mechanobiology Research Assistant**
PI: Dr. Nelly Andarawis-Puri *Cornell University*

- Characterized the role of poorly understood non-collagenous proteins (elastin) in tendon fatigue damage accumulation via mouse models of overuse injuries.
- Developed custom quantitative analysis pipelines to characterize tendon damage and fatigue following cyclic mechanical loading.
- Designed a physical therapy glove for at-home rehabilitation of the hand and developed a mobile app paired with the device to relay information between patient and physician/therapist (senior design project guided by Dr. James Antaki).

Sept 2018 – Dec 2019 **Immunotherapy & Cell Engineering Research Assistant**
PI: Dr. Ankur Singh *Cornell University*

- Engineered microfluidic devices and embedded lymphoma cells in hydrogels to validate the use of 3D human organ culture models for translational cancer research.

May – Aug 2018 **Radiation Oncology Research Intern**
Mentors: Dr. Benjamin Lang, Dr. Stuart Calderwood *Beth Israel Deaconess Medical Center*

- Established data analysis pipelines for gene and protein expression analysis in R.
- Quantified expression of a tumor biomarker (Hsp70) within different subtypes of human breast cancer and examined molecular pathways activated within irradiated tumor cells.

SCIENCE POLICY & COMMUNICATION

Jun 2025 – Current **District Visit Leader & Grassroots Advocacy Team Member**
Remote *Scientist Network for Advancing Policy (SNAP)*

- Organize congressional district engagement across Arizona, Colorado, and Michigan focused on federal investment in science and research.
- Maintain relationships with legislative staff, monitor developments in AI and emerging technologies, and develop evidence-based communications for policy audiences.

Jun – Aug 2026 **Science & Technology Policy Summer Scholar**

Institute for Science & Policy *Denver Museum of Nature & Science*

- Synthesized multidisciplinary scientific literature and stakeholder perspectives into policy materials for Colorado state legislators.
- Researched issues spanning aging, artificial intelligence, healthcare, water, energy, transportation, and education.
- Translated complex scientific and technical information into clear, actionable communications for decision-makers.

PRESENTATIONS

Oral Presentations

2026 "A Plan to Tackle Social Isolation Amongst Older Coloradans." Final Policy Proposal Presentation. *Colorado Science & Technology Policy Program*.

2026 "The Infamous Cancer Cell Hatches a New Evil Plan…to Destroy Bone!" Three Minute Thesis Finalist Talk.

2026 "The Role of Mechanical Signaling in Breast Cancer-Induced Bone Disease." Invited Seminar. *Hernandez Lab, UCSF*.

2025 P.E.O. (Philanthropic Educational Organization) Scholarship Finalist. Invited Research Talk. *P.E.O. Colorado Chapter*.

2025 **Venkatesh S***, et al. "SERPINE1 Underpins the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *ASBMR Annual Meeting*. Podium. [awarded]

2025 "Advanced Academic Development: Manifestation and Visualization as Tools for Academic Performance Planning." University of Colorado Boulder. Guest lecture for Dr. Leslie Blood.

2025 **Venkatesh S***, et al. "Characterizing the biochemical responses that upset bone homeostasis in the metastatic vicious cycle." *Advances in Mineral Metabolism (AIMM) ASBMR John Haddad Young Investigator Meeting*. Late-Breaking Talk.

2025 Session Moderator (Co-Chair). "Oncology Meets Osteology: Advances in Cancer & Bone Research." *AIMM ASBMR John Haddad Young Investigator Meeting*.

2024 **Venkatesh S***, et al. "Dynamic Flow Alters the Breast Cancer Cell Mechanoresponse in a 3D Perfusion Model of Bone Metastasis." *MechBio Symposium*. Podium. [awarded]

2023 Hayes A*, Sundaram V, Williams G, **Venkatesh S***, et al. "Meeting Schools Where They Are: Integrating Engineering Outreach Curriculum in the Classroom Without Forcing an Agenda." *American Society for Engineering Education: RMS Conference*. Podium.

2019 "And in This Corner: Hsp70 and Breast Cancer." *CURBx Talks*. Public talks in the style of TEDx. [awarded]

Poster Presentations

2026 "Mechanical Forces and Aging: A Vicious Cycle of Bone Destruction by Breast Cancer Metastasis." *Achievement Rewards for College Scientists (ARCS) Colorado Spring Gala*.

2026 **Venkatesh S***, et al. "SERPINE1 is Overlooked in the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *AAAS Annual Meeting*. [awarded]

2026 **Venkatesh S***, et al. "SERPINE1 Underpins the Tumor-Osteocyte Response to Fluid Shear and Irradiation." *Bones and Teeth Gordon Research Conference*.

2025 **Venkatesh S***, et al. "Dual Role of Periostin in Mediating the Bone Mechanoresponse in Metastasis In Vivo." *ASBMR Annual Meeting*.

2025 "Characterizing the Vicious Cycle of Bone Destruction by Breast Cancer Metastasis." *ARCS Colorado Spring Gala*.

2024 **Venkatesh S***, et al. "Dynamic Flow Alters the Breast Cancer Cell Mechanoresponse in a 3D Perfusion Model of Bone Metastasis." *MechBio Symposium*. [awarded]

2023 **Venkatesh S***, et al. "Dynamic Fluid Flow Alters Breast Cancer Cell Mechanoresponse in a 3D Perfusion Model of Bone Metastasis." *Cancer and Bone Society Meeting*.

2021 **Venkatesh S***, Chatterjee M, Andarawis-Puri N. "Enzymatic Depletion Highlights Elastin's Potential Role in Tendon Fatigue Damage." Cornell Undergraduate Research Board, Spring Symposium.

2021 Chatterjee M*, Acosta A, Gaur S, **Venkatesh S**, Taub PJ, Andarawis-Puri N. “MRL/MpJ Exhibit Restored Function and Insertion Site Structure in Supraspinatus Tendon after Surgical Repair.” *ORS 2021 Annual Meeting*.

2020 Chatterjee M, **Venkatesh S***, Liu EJ, Andarawis-Puri N. "Unaltered Collagen Organization Following Rotator Cuff Overuse Injury Suggests Damage to Other Microstructural Components.” *2020 BMES Annual Meeting*.

2019 Lang BJ, Holton KM, Guerrero ME, **Venkatesh S***, et al. “Hsp70 Advances Mammary Tumor Metastasis by Associated Matrisome Gene Upregulation.” *CURB Spring Symposium*.

*Presenting author.

TEACHING, OUTREACH, AND SERVICE

2022 – 2026 **Court-Appointed Special Advocate**, *Boulder County*

Advocated for children involved in dependency and neglect proceedings, collaborating with families, caseworkers, attorneys, educators, and other stakeholders and preparing reports for the court.

2021 – 2026 **Outreach Program Lead**, *University of Colorado Boulder*

Led K–12 science and engineering outreach, collaborating with educators to develop programming, recruit volunteers, and organize student experiences.

2022 – 2025 **Graduate Teaching Assistant**, *University of Colorado Boulder*

Courses: Statics and Structures, Dynamics, Mechanics of Solids, Mechanics of Cancer, Introduction to Engineering Computing.

2022 – 2024 **Graduate Student Representative**, *University of Colorado Boulder*

2021 – 2023 **Project-Based Learning Mentor**, *University of Colorado Boulder*

2018 – 2021 **Undergraduate Teaching Assistant**, *Cornell University*

Courses: Systems Mechanobiology, Biomedical Instrumentation, Investigative Biology Laboratory.

2018 – 2021 **Children’s Science Program Volunteer**, *Ithaca Sciencenter*

SELECTED HONORS & AWARDS

2026 Carried University of Colorado Boulder Mace in the traditional commencement procession (nominated by faculty).

2026 Mechanical Engineering Graduation Ceremony Student Speaker (nominated by faculty and staff).

2026 First Place. Technology, Engineering & Math Category, *AAAS E-Poster Competition*.

2026 Three Minute Thesis (3MT) Finalist.

2025 Young Investigator Travel Grant, *ASBMR Annual Meeting*.

2024 – 2026 Graduate Research Scholar, *ARCS Foundation*.

2024 Best Graduate Scientific Brief & Poster Presentation, *MechBio Symposium*.

2024 Committee for Equity in Mechanical Engineering Doctoral Service Award, *University of Colorado Boulder*.

2022 – 2023 Community Engagement & Outreach Awards, *University of Colorado Boulder*.

2023 AAAS CASE Workshop Sponsorship (one of two *Boulder* students).

2023 Future K-12 Leaders Service Award, *University of Colorado Boulder*.

2023 Visiting Research Scholar, *University of Massachusetts Amherst*.

2022 GRFP Honorable Mention, *National Science Foundation*.

2021 Chair’s Graduate Fellowship, *University of Colorado Boulder*.

2021 Best Science Communication and Applied Research Finalist, *CURB Spring Symposium*.

2019 People's Choice Award, *'CURBx' Talks*.

2017 John McMullen Dean's Scholar, *Cornell University*.

SKILLS

***Software*:** MATLAB. R. Arduino. Java. Python. Microsoft Excel. Excel Solver. Autodesk Fusion 360. SolidWorks. APM Planner. QGroundControl. Cura. OpenSim. CircAdapt. SnapGene. GraphPad. ImageJ. Image Studio Lite. ANSYS. ABAQUS. Simpleware ScanIP. OpenFOAM. DragonFly. IVIS Lumina. Supercomputing Clusters. Bash (Unix shell command-line interpreter).

***Mechanical and Electrical*:** Mechanical Device Design. Wiring. CAD. 3D Printing. Laser Cutting. Soldering. Circuit Design. Oscilloscope. PDMS Microfluidic Device Fabrication. Tensile Testing. Dynamic/Fatigue Loading. Multiaxial Ball Burst Testing. Perfusion & Compression Bioreactor Systems.

***Chemical and Biological*:** Cell & Tissue Culture. Cell Cryopreservation. Western Blot. Real-Time qPCR. Histology. Flow Cytometry. Organoid Generation. Boyden Chamber Assays. RNA Purification. Restriction Enzyme Digests. Gel Electrophoresis. Bacterial Transformation & Transfection. BCA & Ninhydrin-Based Assays. ELISA. Immunohistochemistry. Hydrogel & Scaffold Fabrication. Next-Generation Sequencing (RNA-Seq, sample/library prep). Ex Vivo Cultures. Raman Spectroscopy.

***Other Technical Skills*:** Digital Image Correlation (DIC). Mouse Husbandry. Modern Regression Analysis. Nonparametric Statistical Methods. Fluorescence Microscopy. 2D, 3D Microscopy. Spectrophotometry. Second Harmonic Generation (SHG) Imaging. Basic App Development.

Professional: Scientific Writing. Literature Synthesis. Grant Writing. Science Communication. Policy Analysis. Stakeholder Engagement. Public Speaking.

MEDIA

2026 "Meet 3MT Finalist Shreya Venkatesh." University of Colorado Boulder, Graduate School.

2025 "Fight for continuation of scientific funding." *Scottsdale Progress*, Letters to the Editor. McClintock Letters Initiative, *SNAP Coalition*.

2023 "Researchers at CU Boulder land NSF Award for air and soil quality education."

2023 "CU Boulder students get high schoolers excited about STEM fields through outreach event."